Néstor García

Mechanical Construction Manager

International site execution leader across Oil & Gas, Petrochemical, Power Generation, Renewables and Biofuels — performing roles from QA/QC Manager to Site Manager and Client Representative across Europe, Africa, the Middle East and North America.

BASED
Valencia, Spain

STATUS
Available for new projects

Oil & Gas | Petrochemical | Power Generation | Renewables & Biofuels | Recycling | Mission-Critical / Hyperscale

nestorgarcia.emp@gmail.com | +34 627 225 615 | linkedin.com/in/ngarcia-mechanical-construction-manager | nestorgarcia-mcm.github.io

34	23.3 yrs	2001–2026	50%	19	17
total projects	years of experience	career span	of projects Greenfield	projects as final warranty stamp responsible	complex revamp & retrofit projects

Profile

Mechanical Construction Manager with solid international experience in industrial and critical infrastructure projects — Oil & Gas, Petrochemical, Power Generation, Renewables and Recycling — performing key roles such as Mechanical Construction Manager, Client Representative, Senior Mechanical Supervisor, QA/QC Manager and Site Manager, leading Greenfield and Brownfield projects in Europe, Africa, the Middle East and North America.

Proven leader in site execution, vendor surveillance, QA/QC, contractor management, supervision, installation and commissioning of mechanical systems (refrigeration, process, chemical separation, water treatment, firefighting, storage and others), guaranteeing quality, safety and compliance with deadlines in critical environments, with advanced capabilities in databases, Power BI, SQL, Excel, digital construction workflows and AI-assisted reporting.

Accustomed to working under strict construction standards in critical projects. Demonstrated ability to lead multicultural teams, solve problems in the field, and ensure safe, efficient and on-time delivery in hyperscale projects.

Recent Experience

2025 – AUG 2026

Senior Mechanical Construction Manager

Aldesa Industrial — Spain

Construction of 2 units for Moeve at the new 2G biofuel plant in Huelva, set to be the largest 2G biofuel complex in southern Europe. Prefabrication and installation of 250K pipe-welding inches, structural and static/rotating equipment, insulation, and Electrical & Instrumentation (E&I). Power BI model crossing 35,017 engineering and SAP material codes to forecast welding inches before material arrived.

Education

B.Sc. Computer Databases IT
Valencia Polytechnic University, Spain — 2001

Graduate Certificate of Education (GCE)
I.B. Ramon Llull, Valencia, Spain — 1997

Certifications

Inspector of Welded Constructions, Level II	07/2016
CFR — Certified First Responder	03/2020
Mass Transfer Technologies	08/2023

Experience

2023 – 2025

Project Manager / Senior Mechanical Technician

Sulzer Chemtech, Tower Field Services — Europe · Africa · Middle East

Tower Field Services representative and specialist technician, certifying works and advising the client through to warranty sign-off across multiple projects under the Total Tower Care model.

SERVICES DELIVERED

- Mass transfer internals — installation and replacement of conventional, high-performance and cartridge trays, structured and random packings, distributors, collectors and mist eliminators.
- Revamps & retrofits — mass transfer equipment replacement and column revamps, including mechanical management and QA/QC of a crude tower revamp, with milestones completed ahead of schedule.
- Turnarounds & shutdowns — pre-shutdown engineering, turnaround schedule planning and vessel integrity restoration.
- Emergency response — emergency replacement of column internals and emergency plant repairs on static equipment.
- Automated weld overlay — corrosion and erosion protection to extend vessel service life.
- Engineering & technical support — feasibility and debottlenecking studies to improve column performance.

RESPONSIBILITIES & RESULTS

- Certified completed works and advised the client from execution to final handover.
- Signed off 5-year warranties on completed weld overlay works and repairs.
- Managed simultaneous projects across Europe, Africa and the Middle East.
- Ensured full compliance with API and ASME standards, meeting all safety and quality KPIs.
- Increased client satisfaction by 50% by delivering projects weeks ahead of schedule.

2022 – 2023

Mechanical Construction Manager

Técnicas Reunidas — Albacete, Spain

Mechanical construction management of the Repetco Innovations PET/PE recycling plant, a highly complex industrial facility built around self-contained mechanical package units integrated into the overall process, with detailed design and mechanical interconnection services coordinated by Técnicas Reunidas. Identified and resolved construction issues on-site to prevent delays; monitored subcontractor operations and applied time, resource and cost-management strategies to deliver under-budget completion.

MECHANICAL PACKAGES

- Main process packages (plastic treatment) — crushing and milling units reducing complex post-consumer trays and containers to a homogeneous flake size; patented mechanical delamination modules that separate the PE layer from the PET using controlled friction, temperature and agitation, with no polluting solvents; washing lines and density-separation tanks or hydrocyclones for the separated polymers.

- Auxiliary services & heat exchange — pressure vessels and heat exchangers managing the water and thermal fluid flows that hold the temperatures needed to weaken the film's adhesives; storage silos with dosing, discharge and pneumatic conveying of granulated material between process stages; industrial HVAC and ventilation for air quality and emissions control in the processing halls.
- Protection & fluids — fire protection systems (pumping networks and water storage around the plant perimeter, to meet plastic-waste safety regulations); process water treatment modules (filtration and mechanical purification) that recirculate wash water and minimise water consumption.

Core Responsibilities

- Site leadership & delivery — lead multidisciplinary site teams, subcontractors and vendors to deliver mechanical scope on schedule and to quality standards.
- Mechanical execution — piping systems, HP/MP piping, static and rotating equipment, pressure vessels, mass transfer columns, boilers, heat recovery and refrigeration systems (chillers, CRAH/CRAC, pumps, exchangers, cooling towers, firefighting, CHW/CWS networks, water treatment and auxiliary equipment).
- QA/QC — weld inspection, support & alignment checks, pressure testing, flushing, chemical cleaning and material verification.
- Daily coordination — with contractors, engineering, HSE, QA/QC and commissioning to ensure correct execution of MEP systems.
- Technical review — of P&IDs, isometrics, assembly drawings, ITPs, WPS/PQR, installation procedures and test methods.
- Management — of NCRs, punch lists, RFIs, technical deviations and field interference resolution.
- Commissioning support — functional testing, start-ups, hydraulic balancing, ΔT validation and thermal performance.
- Planning & progress control — logistics and monitoring of mechanical milestones on site.
- Vendor surveillance — in international workshops for mechanical equipment, ensuring compliance with ASME, EN and customer specifications.
- Technical representation — in front of international clients, engineering teams and suppliers.

Technical & Digital Skills

Construction & Engineering

Full mechanical package — piping, rotating & static equipment, high-pressure components, skid systems, pressure vessels.

Cooling & Mission-Critical MEP

Chilled water systems, condenser cooling, heat rejection systems, redundancy strategies (N+1 / N+2) and commissioning protocols for mission-critical uptime.

Digital Construction & BIM

Practical integration of Revit and Navisworks into large databases — parameter extraction, ETL workflows, 4D/5D coordination and Power BI visualization to support prefabrication and clash mitigation.

Contract & Cost Control

Cost baseline development, earned value concepts for mechanical scope, change-order workflows, commercial risk registers and mitigation plans, SQL-backed cost reconciliation and variance reporting.

Process, Data & AI Innovation

Digitalization of inspections, material & equipment tracking systems, integration of construction data into centralized platforms, and development of proprietary progress / productivity / risk-control systems connecting contractors, engineering, QA/QC and commissioning. Self-built SQL project database (34 projects, about 140,000 documents) connected to Power BI and published as an online portfolio (nestorgarcia-mcm.github.io). AI assistants (Claude, ChatGPT, Microsoft 365 Copilot) for specification, datasheet and contract review, correspondence triage and reporting, with reusable instructions, scheduled agents and human sign-off.

Data, Digital & AI

Power BI
Dynamic dashboards — MTO vs. progress, QA/QC heatmaps, punch-list status, cost vs. progress.

Oracle, SQL & MS Access
Self-built SQL project database (34 projects, ~140,000 documents) feeding Power BI and nestorgarcia-mcm.github.io.

AI assistants & agents
Claude (skills, connectors, scheduled agents), ChatGPT and M365 Copilot for document review, triage and reporting.

Excel (advanced)
Power Query, VBA, pivot modelling and Python data scripts (AI-assisted) for reconciliations and automation.

Materials & Codes

Standards & specifications

ASME I / II / V / VIII / IX | ASME B31.1 / B31.3 / B31.4 / B31.8 | API | UNE-EN | EN | ISO 9001 / 14001 | DIN EN | ASHRAE | EN 378 | RITE

Alloy materials, high pressure & temperature

Carbon Steel | Stainless Steel | Alloy Steels | Inconel | P22 | P91

Languages

Spanish	**Bilingual / Proficient (C2)**
English	**Advanced (C1)**
French	**Intermediate (B1)**
Portuguese	**Intermediate (B1)**
Italian	**Elementary (A2)**